Residual certificates for the Erdős–Straus conjecture: exact solvability criteria, a sieve bound, and computations to 10^{11}

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Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/a+1/b+1/c$ is solvable in positive integers for every $n \geq 2$; the open cases concentrate on primes in six residue classes mod 840 (*hard primes*). We develop the *residual method*: a solution with first denominator $a = (p+R)/4$ exists if and only if some divisor $k \mid m^2$, $m = pa$, satisfies $k \equiv -m \pmod{R}$, and every solution arises this way. Solvability at any fixed residual — prime or composite — is a computable function of finite data, yielding exact criteria at $R = 3, 7, 11, 15$, and a reciprocity identity $\left(\frac{q}{R}\right) = \left(\frac{p}{q}\right)$ governs how failures correlate across residuals. Failure at R forbids at least half the unit classes mod R on the shifted form $(p+R)/4$ — proved for every residual at once via Kneser's addition theorem — giving exceptional sets $O_{\mathcal{P}}(x/(\log x)^{1+|\mathcal{P}|/2})$ for every finite admissible residual list $\mathcal{P}$ drawn from the primes and the composites up to 107 (marginally weaker exponents beyond): weaker in density than Vaughan's classical $x\exp(-c(\log x)^{2/3})$, but explicit and certificate-producing, with the elementary layer and the core finite enumerations of the proofs machine-checked in Lean 4/mathlib. Letting the residual list grow with x yields an unconditional almost-all theorem: all but $O(x\exp(-c(\log\log x)^2))$ hard primes $p \leq x$ satisfy $R_{\min}(p) \leq \varepsilon \log\log x$. A reciprocity-selected *ladder* of p-adapted residuals then localizes the conjecture for hard primes into a single measured counting inequality, with the first rung proved and a matching half-dimensional lower bound for true failures given in outline. Computationally we exhibit certificates for all 128,671,219 hard primes below 10^{11}; the record minimal residual is 107.

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1 Introduction

Erdős and Straus conjectured (see [5]) that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{1.1}$$

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has a solution in positive integers a, b, c. By multiplicativity it suffices to treat prime n, and classical polynomial identities settle every prime outside the six residue classes

$$\mathcal{H} \bmod 840 \;=\; \{1, 121, 169, 289, 361, 529\},$$

the squares of units modulo 840 (Mordell [12, Ch. 30]; the quadratic-residue nature of the obstruction goes back to Yamamoto [19]). We call primes in these classes *hard primes*. Direct verification has confirmed the conjecture far beyond any range considered here: to 10^{14} by Swett [15], to 10^{17} by Salez [13], and to 10^{18} by Mihnea and Bogdan [11]; the present paper is logically independent of those bounds. The structure of the solution count was analyzed by Elsholtz and Tao [4]. Verification, however, carries no structural information; the aim here is a quantitative structural reduction. For calibration: the first density bounds on the exceptional set are due to Webb [18], sieve bounds were advanced by Sander [14], and Vaughan's classical mean-value bound [17] gives an exceptional set $\ll x\exp(-c(\log x)^{2/3})$ for (1.1) — asymptotically stronger than any fixed power of $\log x$; the value of the present approach is not raw density but its explicit, certificate-generating, machine-verified structure, together with the exact criteria.

New contributions

- The *residual formulation* of the Erdős–Straus equation: solvability with first denominator $(p+R)/4$ is a divisor-class condition, complete over all solutions (Proposition 1.1).
- *Exact* solvability criteria at $R = 3, 7, 11$ (Theorems 1.2, 1.4, 1.5) and at the first composite residual $R = 15$ (Theorem 1.6: a Jacobi-character dichotomy with no budget cases), and a meta-theorem (Theorem 1.3) showing every fixed residual — prime or composite — admits a finite computable exact criterion.
- A reciprocity principle (Theorem 1.8) collapsing the per-residual conditions into one global statement about p — which we read as the organizing principle behind correlated failure and the record prime.
- The per-residual sieve input proved unconditionally for *every* residual at once (Theorem 1.10): via Kneser's addition theorem, any failing configuration occupies at most $(R-3)/2$ nonzero factor classes — closing what was previously a per-residual machine verification capped at $R = 107$ (Lemma 1.9, now an independent confirmation), and extending to composite residuals through the general abelian form.
- A constructive sieve chain built on these criteria (Theorem 1.11, Proposition 1.12): explicit, certificate-producing exceptional-set bounds of exponent $1 + |\mathcal{P}|/2$ for every finite admissible list $\mathcal{P}$ of primes and of composites up to 107, with marginally smaller exponents for larger composites with $\omega(R) \le 2$ (29/2 on the 27 residuals up to 107; 31/2 with the two identity families of Proposition 1.12 adjoined); an unconditional density reduction (Theorem 1.13): for every A, a finite residual set whose joint failure set is $O_A(x/(\log x)^{1+A})$; and the *uniform chain* (Theorem 4.5), the strongest unconditional statement here: all but $O(xe^{-c(\log\log x)^2})$ hard primes $p \le x$ have $R_{\min}(p) \le \varepsilon \log\log x$, with the three walls that make $\exp(-c(\log\log x)^2)$ the ceiling of the method identified (Remark 4.6).
- Computed and verified residual certificates for all 128,671,219 hard primes below 10^{11}, with a calibrated model whose deep-tail predictions were subsequently confirmed (Section 6).

- The *Burgess–reciprocity ladder* (Section 5): a p-adapted residual family on which the all-residue failure mode is impossible by construction; the ladder chain and its first rung proved (Theorems 5.3, 5.4), the avoidance ladder two-sided and true failures bounded below via the half-dimensional sieve (Theorems 5.5, 5.7; established in outline, with proof sketches), and an almost-all bound past every fixed power of $\log x$ conditional on one measured hypothesis (Theorem 5.9).

- A machine-checked (Lean 4 [3] with mathlib [10]) formalization, in the accompanying repository, of the elementary layer of the proofs, of the finite enumerations behind Theorems 1.4 and 1.5 and Lemma 1.9 at $R = 19, 23, 31$, and of the reduction of Theorem 1.3 itself. The complete inventory, and what each layer contributes to the trusted base, is in Appendix A.

Conceptually, the paper has three layers: (1) an exact reformulation of solvability — every solution of (1.1) at a hard prime is a residual certificate (Proposition 1.1); (2) finite, machine-checkable inputs for individual residuals — the exact criteria (Theorems 1.2–1.6), the reciprocity identity (Theorem 1.8), and the support bounds (Lemma 1.9, Theorem 1.10); and (3) sieve consequences obtained by combining those inputs (Theorems 1.11–1.13), capped by the p-adapted ladder program of Section 5. The computations of Section 6 calibrate layer (3) and test its predictions; the formal development machine-checks the core of layers (1)–(2).

1.1 The residual method

Fix a hard prime p. All six classes satisfy

$$\text{(H1) } p \equiv 1 \pmod 8, \qquad \text{(H2) } p \equiv 1 \pmod 3, \qquad \text{(H3) } p \bmod 7 \in \{1, 2, 4\}.$$

For an *admissible residual* $R \equiv 3 \pmod 4$ put

$$a \;=\; \frac{p+R}{4} \in \mathbb{Z}, \qquad m \;=\; pa.$$

Substituting a into (1.1) and clearing denominators shows that a solution with first denominator a exists if and only if

$$\exists\, k \mid m^2 \quad \text{with} \quad k \equiv -m \pmod R, \tag{1.2}$$

in which case $b = (k+m)/R$ and $c = (m^2/k+m)/R$ are positive integers solving (1.1). We write $R_{\min}(p)$ for the least admissible R for which (1.2) holds. The formulation loses no solutions:

Proposition 1.1 (completeness)**.** *Every solution of* (1.1) *at a hard prime p, ordered $a \le b \le c$, has $p/4 < a \le 3p/4$; hence $R = 4a - p$ is an admissible residual in* $(0, 2p]$ *and the solution is a residual certificate at R. Consequently $R_{\min}(p) \le 2p$ if and only if the conjecture holds at p.*

Proof. With $a \le b \le c$, the equation gives $1/a < 4/p \le 3/a$, so $p/4 < a \le 3p/4$ and $R = 4a - p \in (0, 2p]$; $R \equiv 3 \pmod 4$ is automatic from $p \equiv 1 \pmod 4$. By the displayed equivalence, the solution yields a divisor k as in (1.2), so $R_{\min}(p) \le R \le 2p$. Conversely (1.2) at any admissible R produces a solution. □

Two elementary congruences drive everything below. From $4a = p + R$,

$$a \equiv 4^{-1}p, \qquad m \equiv 4^{-1}p^2 \pmod R, \tag{1.3}$$

and since $\gcd(p,R) = 1$ (as p is prime and $R \neq p$, $R \equiv 3 \not\equiv p \pmod 4$), also $\gcd(a,R) = \gcd(m,R) = 1$: the setting is always a unit computation modulo R. By (1.3), m is a *square* modulo every prime $r \mid R$; hence for every prime $r \mid R$ with $r \equiv 3 \pmod 4$ (at least one exists, since $R \equiv 3 \pmod 4$),

$$\left(\tfrac{-m}{r}\right) = \left(\tfrac{-1}{r}\right) = -1 \tag{1.4}$$

— *the target class $-m$ is a quadratic non-residue mod r.*

A worked example. Take the hard prime $p = 1009$ ($\equiv 169 \pmod{840}$) and the smallest admissible residual $R = 3$. Then $a = (1009+3)/4 = 253 = 11 \cdot 23$ and $m = pa = 255{,}277$. Since $11 \equiv 2 \pmod 3$, Theorem 1.2 below promises a certificate: take $k = 11$, a divisor of m^2 with $k \equiv -m \equiv 2 \pmod 3$. Then $b = (k+m)/3 = 85{,}096$ and $c = (m^2/k + m)/3 = 1{,}974{,}822{,}872$, and indeed

$$\frac{4}{1009} = \frac{1}{253} + \frac{1}{85{,}096} + \frac{1}{1{,}974{,}822{,}872}.$$

Had every prime factor of a been $\equiv 1 \pmod 3$, every divisor of m^2 would be $\equiv 1 \pmod 3$ and the required class 2 would be unreachable — the prototype of all the obstructions studied below.

The shape of what follows. Everything in this paper grows out of the picture just seen, read at three levels. *Locally* (one residual): whether the target class $-m$ is attainable from the factor classes of a is a question about a finite multiplicative state — the multiset of classes with bounded exponent budgets — so each fixed R admits an exact, machine-enumerable solvability criterion (Theorems 1.2–1.5 below). *Globally* (all residuals at once): the conditions entering these criteria are quadratic characters, and reciprocity folds the condition at every residual into signs $\left(\frac{p}{q}\right)$ indexed by the primes q dividing the shifted values $(p+R)/4$ — failures at different residuals correlate because shifted values share prime factors (Theorem 1.8). *Quantitatively*: each criterion says that failure forces $(p+R)/4$ to avoid prime factors in many residue classes, a sifting condition; stacking residuals multiplies the savings (Theorem 1.11), though it provably cannot reach emptiness (Remark 4.3) — the honest endpoint of the method, discussed in Section 7.

1.2 Main results

Theorem 1.2 ($R = 3$; exact)**.** *Let $p \equiv 1 \pmod 4$ be prime with $3 \nmid p$, and $a = (p+3)/4$. Then* (1.2) *holds for $R = 3$ if and only if a has a prime factor $q \equiv 2 \pmod 3$. When it does, $k = q$ is a certificate.*

Modulo 3, character equals class and the criterion is elementary. For every other fixed residual, the reduction to a finite check is itself a theorem, and the next two results are its first showcase instances.

Theorem 1.3 (meta-theorem)**.** *For every fixed prime $R \equiv 3 \pmod 4$, the validity of* (1.2) *at a hard prime p depends only on* (i) *the multiset of classes mod R of the prime factors of $a = (p+R)/4$, with multiplicities capped at $\lceil (R-1)/2 \rceil$, and* (ii) *$p \bmod R$. Consequently an exact solvability criterion exists for every R and is computable by finite enumeration.*

Theorem 1.4 ($R = 7$, hard primes; exact)**.** *Let p be a hard prime and $a = (p+7)/4$. Then* (1.2) *holds for $R = 7$ if and only if a has a prime factor that is a quadratic non-residue modulo 7.*

Theorem 1.5 ($R = 11$, hard primes; exact)**.** *Let p be a hard prime and $a = (p + 11)/4$; note $3 \mid a$, $2 \nmid a$, $7 \nmid a$ automatically. Then* (1.2) fails *for $R = 11$ if and only if either*

(a) every prime factor of a is a quadratic residue mod 11; or

(b) $v_3(a) = 1$, every other prime factor of a is $\equiv 1 \pmod{11}$ except a non-residue part *— its factors in the non-residue classes mod 11 — of exactly one of the shapes*

$$\{q \equiv 2\}, \qquad \{q \equiv 6\}, \qquad \{q \equiv 2\} \cup \{q' \equiv 6\} \pmod{11},$$

each with multiplicity one, paired respectively with $p \equiv 2,\ 6,\ 1 \pmod{11}$.

For composite R the certificate condition couples the prime-power components of R through the Chinese Remainder Theorem, and the divisor-class computation lives in the (generally non-cyclic) unit group $(\mathbb{Z}/R)^*$. Theorem 1.3 holds verbatim with $(\mathbb{Z}/R)^*$ in place of the cyclic group (multiplicities capped once exponent ranges cover each class's cyclic span; see Remark 3.1). The first composite case turns out to be *simpler* than $R = 11$: a pure character dichotomy, with the quadratic character modulo a prime replaced by the Jacobi character modulo 15, and no budget cases at all.

Theorem 1.6 ($R = 15$, hard primes; exact)**.** *Let p be a hard prime and $a = (p + 15)/4$; note $p \bmod 15 \in \{1, 4\}$ and $2 \mid a$ automatically. Then* (1.2) *holds for $R = 15$ if and only if a has a prime factor q with $\left(\frac{q}{15}\right) = -1$, i.e. $q \equiv 7, 11, 13, 14 \pmod{15}$.*

Remark 1.7 (computer-assisted components)*.* The exhaustive verifications behind Theorems 1.4, 1.5, 1.6 and Lemma 1.9 below are *parts of the respective proofs*, on the same footing as a hand case analysis. Each is a deterministic enumeration of a finite configuration space whose finiteness and sufficiency are proved in advance (Theorem 1.3; for Lemma 1.9, monotonicity of target-reachability in support and multiplicities, checked at minimal exponents). The checks are short (seconds each, up to 331 s for the largest, $R = 107$), involve only exact integer arithmetic, and the code is available in the accompanying repository; Table 5 records the exact configuration counts and entry points. The enumerations at $R = 7, 11$ and the $R = 19, 23, 31$ cases of Lemma 1.9 are additionally formalized in Lean 4, with the reductions they rely on proved as theorems rather than trusted (the $R = 15$ enumeration and the remaining cases of Lemma 1.9 run in the Python layer only); Appendix A gives the inventory and the exact trust base.

The obstruction framework of Section 2 yields a reciprocity identity that converts the per-residual conditions into a single global statement about p — a structural interpretation of joint failure across residuals:

Theorem 1.8 (reciprocity structure)**.** *Let p be a hard prime, $R \equiv 3 \pmod 4$ prime, and q an odd prime factor of $a = (p + R)/4$ (necessarily $q \neq R$). Then*

$$\left(\frac{q}{R}\right) \;=\; \left(\frac{p}{q}\right).$$

The sieve consequences below need one input per residual: a statement that failure at R forces the shifted form $(p + R)/4$ to avoid prime factors in a positive-density set of residue classes. For $R = 3, 7, 11, 15$ the exact criteria provide this; for every other residual it is the support bound, which we state first in the per-residual machine-verified form in which it was originally established (Lemma 1.9) and then prove unconditionally for every residual at once (Theorem 1.10; both proved in Section 4).

Lemma 1.9 (support bound; verified for every prime $R \equiv 3 \pmod 4$, $19 \le R \le 107$)**.** *Let R be one of the twelve primes $19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 103, 107$, and call the set of unit classes mod R occupied by the prime factors of $(p+R)/4$ its* support*. Every configuration for which residual R fails at a hard prime has support of size at most $(R-1)/2$, at most $(R-3)/2$ of whose classes differ from the identity class (which never affects solvability). Consequently failure at R forbids at least half the classes, through finitely many maximal-support branches.*

Theorem 1.10 (unconditional support bound)**.** (i) *Let $d \ge 4$ be even and $S \subseteq \mathbb{Z}/d \setminus \{0\}$. If the bounded subset-sum set*

$$M(S) \;=\; \Big\{ \textstyle\sum_{v \in S} e_v v \;:\; e_v \in \{0,1,2\} \Big\}$$

is not all of $\mathbb{Z}/d$, then $|S| \le d/2 - 1$, and the bound is attained ($S =$ the nonzero even classes). (ii) *Consequently, for* every *prime $R \equiv 3 \pmod 4$ and every p: if the prime factors of $(p+R)/4$ occupy at least $(R-1)/2$ nonzero unit classes mod R, then every unit class is a divisor class of m^2 and (1.2) holds; failure forces the conclusion of Lemma 1.9, with no upper limit on R.* (iii) *In a finite abelian group G of even order g with t involutions (multiplicative form, $M(S) = \prod_{v \in S}\{1, v, v^2\}$): if $M(S) \ne G$ then $|S| \le \max(\lfloor (g+t-2)/2 \rfloor,\, g/2 - 1)$. For $G = (\mathbb{Z}/R)^*$ with $\omega(R) \le 2$ this is $\le g/2$, so failure at any admissible composite R forbids at least $\varphi(R)/2 - 1$ unit classes on $(p+R)/4$ — sieve dimension $\ge \frac{1}{2} - 1/\varphi(R)$. For the admissible composites $R \le 107$ an exhaustive enumeration (`kneser_support_general`, a proof component of this clause) sharpens the maximal failing support to $g/2-1$, so those residuals forbid at least $\varphi(R)/2$ classes and join the chain with full dimension $\ge \frac{1}{2}$ each. For arbitrary $\omega(R)$ the same bound reads $|S| \le \max(\lfloor (g+t-2)/2 \rfloor,\, g/2-1)$ with $t = 2^{\omega(R)} - 1$, so failure forbids at least $(g-t)/2 - 1 \ge \varphi(R)/2 - 2^{\omega(R)-1} - 1$ unit classes — dimension $\ge \frac{1}{2} - (2^{\omega(R)-1}+1)/\varphi(R)$, and since $2^{\omega(R)} = R^{o(1)}$ this is ≥ 0.44 for every admissible R beyond an absolute bound:* every *residual, with no restriction on ω, is a usable sieve input.*

Theorem 1.11 (the chain)**.** *The number of hard primes $p \le x$ failing every residual in $\{3,7\}$, in $\{3,7,11\}$, in $\{3,7,11,15\}$, and in $\{3,7,11,15,19\}$ is, respectively,*

$$O\Big(\frac{x}{(\log x)^2}\Big), \qquad O\Big(\frac{x}{(\log x)^{5/2}}\Big), \qquad O\Big(\frac{x}{(\log x)^3}\Big), \qquad O\Big(\frac{x}{(\log x)^{7/2}}\Big).$$

More generally, by Theorem 1.10, for every *finite set $\mathcal{P}$ of admissible residuals $R \equiv 3 \pmod 4$ — prime, or composite with $\omega(R) \le 2$ — the number of hard primes failing every residual in $\mathcal{P}$ is $O_{\mathcal{P}}\Big(x/(\log x)^{1+\kappa(\mathcal{P})}\Big)$, where $\kappa(\mathcal{P}) = \sum_{R \in \mathcal{P}} \delta_R$ with $\delta_R = \frac{1}{2}$ for R prime or composite ≤ 107 and $\delta_R = \frac{1}{2} - 1/\varphi(R)$ for composite $R > 107$ (the two tiers of Theorem 1.10(iii)); in particular $\kappa(\mathcal{P}) = |\mathcal{P}|/2$ for every list drawn from the residuals up to 107. For the fifteen primes $\equiv 3 \pmod 4$ up to 107 the exponent is $17/2$, and for the full 27-element admissible list $\{3, 7, 11, \ldots, 107\}$ it is $29/2$. All but a proportion $O((\log x)^{-2})$ of hard primes are solved within $\{3,7,11,15\}$ — one power of $\log x$ less than the counts above, since there are $\asymp x/\log x$ hard primes up to x.*

Proposition 1.12 (aggregate identity families)**.** *Let $p \equiv 1 \pmod 4$ be prime and $R \equiv 3 \pmod 4$. If $R \mid p+1$, then $k = ap^2$ is a certificate for residual R; if $R \mid p+4$, then $k = a^2 p$ is. Hence any prime factor $\equiv 3 \pmod 4$ of $p+1$ or of $p+4$ yields an explicit solution, and no divisor shape $k = a^j p^i$ yields a further p-only condition. Adding the two corresponding linear* *forms to Theorem 1.11: the number of hard primes $p \le x$ failing every prime residual up to* 107 *and both families is $O(x/(\log x)^{19/2})$; with the full* 27*-element admissible list the exponent is* 31/2*.*

Theorem 1.13 (density reduction)**.** *For every $A > 0$ there is a finite admissible set $S(A)$ such that the number of hard primes $p \le x$ for which every residual in $S(A)$ fails is $O_A\Big(x/(\log x)^{1+A}\Big)$.*

Remark 1.14 (relation to Vaughan's bound)*.* Theorem 1.13 and the chain of Theorem 1.11 are *weaker in density* than Vaughan's bound $\ll x\exp(-c(\log x)^{2/3})$ [17], which already exceeds every fixed power of $\log x$. Their interest is orthogonal: the exceptional sets here are cut out by finitely many *explicit* multiplicative conditions on shifted forms, each failure of which is constructively certified by an exhibited divisor k, and the underlying lemmas are exact or machine-verified. Vaughan's method counts solutions on average and exhibits neither certificates nor criteria.

Organization. Proofs are arranged by logical dependency. Section 2 develops the obstruction framework (Propositions 2.1, 2.2) and proves Theorem 1.8, whose Corollary 2.3 depends on the Type-I/II dichotomy introduced there. Section 3 proves Theorem 1.2, then Theorem 1.3, then Theorems 1.4, 1.5 and 1.6, which rest on it. Section 4 proves Theorem 1.11 — base case first, then, after the monotonicity reduction (Lemma 4.1), Lemma 1.9 and its unconditional upgrade Theorem 1.10, the higher exponents — followed by Proposition 1.12, Theorem 1.13, and the uniform chain (Theorem 4.5). Section 5 develops the Burgess–reciprocity ladder on top of Theorem 1.10, with its own ladder census (Table 2).

Table 1 records, for each main result, exactly what its proof consumes and its formal-verification status, so that each layer can be audited independently.

Computations (Section 6) provide verified certificates for all hard primes below 10^{11} and calibrate the base-case constant and decay exponents of the chain.

2 The obstruction framework

Proposition 2.1 (character obstruction)**.** *Let $r \mid R$ be prime with $r \equiv 3 \pmod 4$. If every prime factor q of $m = pa$ satisfies $\left(\frac{q}{r}\right) = +1$, then* (1.2) *fails for R.*

Proof. Every divisor k of m^2 is a product of prime factors of m, so $\left(\frac{k}{r}\right) = +1$. A certificate requires $k \equiv -m \pmod r$, whence $\left(\frac{k}{r}\right) = \left(\frac{-m}{r}\right) = -1$ by (1.4) — impossible. ☐

We call failures produced by Proposition 2.1 *Type I*; failures where m possesses a non-residue prime factor modulo every relevant r, yet no divisor of m^2 attains the exact class $-m \bmod R$, are *Type II.*

Proposition 2.2 (guaranteed-success classes)**.** *Let $t \equiv -4^{-1}p^2 \pmod R$. If a has a prime factor q with $q \equiv t\,p^{-i} \pmod R$ for some $i \in \{0,1,2\}$, then $k = q\,p^i$ is a certificate. In particular, since $t\,p^{-2} \equiv -4^{-1} \pmod R$, any prime factor of a in the fixed class $-4^{-1} \bmod R$ certifies success regardless of p.*

Proof. $k = qp^i$ divides m^2 (as $q \mid a$ and $i \le 2$) and lies in class $t \equiv -m$. Integrality of b, c is automatic from $k \equiv -m$ and $m^2/k \equiv m^2(-m)^{-1} \equiv -m \pmod R$. ☐

The contrapositive of Proposition 2.2 is the sieve input for Theorem 1.13: failure at R forces $(p+R)/4$ to avoid prime factors in up to three explicit classes modulo R.

The next identity converts the per-residual character conditions into a single global statement about p, collapsing joint failure across residuals into one set of quadratic-character conditions.

Proof of Theorem 1.8. From $q \mid p + R$ we get $R \equiv -p \pmod q$. By quadratic reciprocity with $R \equiv 3 \pmod 4$, $\left(\frac{q}{R}\right) = (-1)^{\frac{q-1}{2}}\left(\frac{R}{q}\right) = (-1)^{\frac{q-1}{2}}\left(\frac{-1}{q}\right)\left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)$. □

Corollary 2.3 (forced primes and correlated failure)**.** *(i) The hard classes are the squares mod* 840*, so p is automatically a quadratic residue mod* $3, 5, 7$*, and* $p \equiv 1 \pmod 8$ *handles* $q = 2$ *(which divides a only for* $R \equiv 7 \pmod 8$*), where* $\left(\frac{2}{R}\right) = +1$*): no prime forced into a shifted value* $(p+R)/4$ *by the congruences (H1)–(H3) carries a negative character. (ii) p fails Type-I at every prime residual* $R \le B$ *simultaneously if and only if* $\left(\frac{p}{R}\right) = +1$ *for each such R and* $\left(\frac{p}{q}\right) = +1$ *for every* unforced *odd prime q (one not forced by the congruences of (H1)–(H3)) dividing any of the* shifted values $(p + R)/4$*,* $R \le B$ *— one condition per distinct prime, shared across every residual it divides.*

Proof. Immediate from Theorem 1.8 together with (H1)–(H3) and the quadratic characters of the forced small primes. □

In words, part (i) says that the small primes forced into the shifted values by the Mordell structure never break a Type-I failure — a conceptual reading of why the hard classes are exactly the square classes. By part (ii), failures at distinct residuals correlate through shared prime factors: shifted values with few distinct prime factors leave few independent sign conditions. Section 6 quantifies this for the record prime. Both parts are verified at scale (135,588 instances of Theorem 1.8, 61,438 forced-prime instances, zero violations of either; Appendix A).

3 Exact criteria

3.1 Proof of Theorem 1.2

Proof. Modulo 3 the unit group has order 2, so quadratic character equals class. Since $p^2 \equiv 1 \pmod 3$, (1.3) gives $m \equiv 1$ and the target is $-m \equiv 2 \pmod 3$; also $3 \nmid a$. *Necessity:* if every prime factor of a is $\equiv 1 \pmod 3$, then $a \equiv 1 \pmod 3$, hence $p = 4a - 3 \equiv 1 \pmod 3$ as well, so every prime factor of $m = pa$ is $\equiv 1 \pmod 3$ and every divisor of m^2 is $\equiv 1 \pmod 3$: class 2 is never attained. *Sufficiency:* $k = q \equiv 2 \pmod 3$ divides m^2, and the recovery formulae give integers as in Proposition 2.2. □

3.2 Proof of Theorem 1.3

Proof. A certificate is a divisor $k \mid m^2$ in class $-m$ mod R. The set of classes of divisors of m^2 is determined by the classes of the prime factors of $m = pa$ together with their multiplicities, each factor being usable with exponent at most twice its multiplicity; exponents are only meaningful modulo $\mathrm{ord}_R(\text{class}) \le R - 1$, so multiplicities beyond $\lceil (R-1)/2 \rceil$ are equivalent to that cap (Remark 3.1 illustrates why). The target class and the consistency relation $\prod(\text{classes}) \equiv a \equiv 4^{-1}p \pmod R$ are functions of p mod R by (1.3). Hence solvability is a function of the capped class multiset and p mod R, a finite amount of data. □

Remark 3.1 (why the cap works, and the size of the space)*.* The one step above that is not pure bookkeeping is the multiplicity cap, and a small example shows both why it is harmless and what it buys. Take $R = 7$ and a factor of a in a class x of order 6 (say $x = 3$): a factor of multiplicity μ may enter a divisor of m^2 with any exponent $0, 1, \ldots, 2\mu$, and once $2\mu \ge 6$ these powers already sweep out the full cyclic group $\langle x \rangle$ — raising μ past $\lceil (R-1)/2 \rceil = 3$ enlarges nothing that a divisor subproduct can reach. For classes of smaller order the cap is generous rather than tight (a class of order 2 stabilizes at $\mu = 1$), which is fine: the enumeration ranges over a finite over-approximation. What finiteness buys is concrete: at $R = 7$ the capped space has 4^6 multisets before the structural constraints (i)–(iii) in the proof of Theorem 1.4 below and 1,536 consistent configurations after them; at $R = 11$ it compresses to 25 equivalence states. The enumeration is decidable, not merely heuristic, precisely because the cap and the consistency relation are proved before any code runs.

3.3 Proof of Theorem 1.4

Proof. ($\Rightarrow$) is Proposition 2.1 with $r = R = 7$. ($\Leftarrow$) reduces to a finite verification via Theorem 1.3. Write $(\mathbb{Z}/7)^* \cong \mathbb{Z}/6$ in discrete-log coordinates. Three structural constraints hold at hard primes:

(i) $p \bmod 7 \in \{1, 2, 4\}$, by (H3);

(ii) $2 \mid a$: by (H1), $p+7 \equiv 0 \pmod 8$, so $a = (p+7)/4$ is even — the class of 2 (a residue mod 7) is always present;

(iii) the factor classes multiply to $a \equiv 4^{-1}p \pmod 7$.

Enumerating all class multisets with multiplicities capped at 3 subject to (i)–(iii) yields 1,536 consistent configurations; for each, one checks whether the target class is attained by a bounded-exponent subproduct. The verification (code: `theory.verify_R7_finite`) confirms: the target is attainable precisely when some factor class is a non-residue. □

Remark 3.2. The hard-class hypotheses are essential: for $p = 701 \equiv 1 \pmod 4$ (not hard), $a = 177 = 3 \cdot 59$ has the non-residue factor 3, yet $R = 7$ fails. Constraint (ii) — the forced even factor — is exactly what rescues every would-be Type-II configuration at hard primes. Empirically (Section 6), among 41,421 sampled failures of $R = 7$ at hard primes, *all* are Type I.

3.4 Proof of Theorem 1.5

Proof. By Theorem 1.3 with $R = 11$ (the unit group has order 10): multiplicities cap at 5, and at hard primes $3 \mid a$ (from (H2), $11 \equiv 2 \pmod 3$), while $2 \nmid a$ (from (H1), $p + 11 \equiv 4 \pmod 8$) and $7 \nmid a$ (from (H3), $-11 \bmod 7 = 3 \notin \{1, 2, 4\}$). A dynamic-programming enumeration over the finite configuration space (code: `theory.finite_criterion_dp`) compresses it to 25 equivalence states: 16 succeed, 6 fail with all factor classes quadratic residues (case (a)), and exactly 3 fail with a non-residue present (case (b)); the three failing states decode to the three listed patterns. Two structural facts emerge from the enumeration. First, no state fails by a *proper-subgroup* obstruction: the only subgroup of $\mathbb{Z}/10$ containing an odd element properly is $\{0, 5\}$, and the consistency relation forces p's class to extend it to all of $\mathbb{Z}/10$. Second, the case-(b) failures are pure *budget* failures — the exponent bounds, not the character obstruction, block the target: any of $v_3(a) \geq 2$, a factor in one of the classes $4, 5, 9$, or a second non-residue factor restores solvability. □

3.5 Proof of Theorem 1.6

Proof. ($\Rightarrow$, the Jacobi obstruction.) By (1.3), $m \equiv 4^{-1}p^2 = (2^{-1}p)^2 \pmod{15}$ is the square of a unit, so the Jacobi symbol satisfies $\left(\frac{m}{15}\right) = +1$; and $\left(\frac{-1}{15}\right) = \left(\frac{-1}{3}\right)\left(\frac{-1}{5}\right) = -1$. Hence $\left(\frac{-m}{15}\right) = -1$.

The classes with $\left(\frac{q}{15}\right) = +1$ are exactly $\{1, 2, 4, 8\} = \langle 2\rangle$, a subgroup of $(\mathbb{Z}/15)^*$ containing p's class ($p \bmod 15 \in \{1, 4\}$, since $p \equiv 1 \pmod 3$ by (H2) and $p \equiv \pm 1 \pmod 5$ for all six hard classes). If every prime factor of a lies in $\langle 2\rangle$, then every divisor of m^2 does, and the target $-m \equiv 11 \pmod{15}$ (an equality valid for both p-classes) is unreachable. This composite-modulus form of Proposition 2.1 is strictly stronger than the proposition applied to the single prime $r = 3 \mid 15$.

($\Leftarrow$.) By the composite form of Theorem 1.3 over $(\mathbb{Z}/15)^* \cong \mathbb{Z}/2 \times \mathbb{Z}/4$ (Carmichael exponent $\lambda(15) = 4$, so reachability saturates at multiplicity 2; multiplicities 0..5 capture every pair of saturation state and class product). The structural constraints at hard primes are: $p \bmod 15 \in \{1, 4\}$; $2 \mid a$ (from (H1), $8 \mid p+15$), so the class of 2 always appears; and consistency $\prod(\text{classes}) \equiv a \equiv 4p \pmod{15}$. Enumerating the full space (code: `theory.verify_R15_finite`) yields 349,920 consistent configurations: 349,380 succeed and 540 fail — and *every* failing configuration has all factor classes in $\langle 2\rangle$. In contrast to $R = 11$ there are no budget failures at all: the consistency relation forces $\prod \left(\frac{q}{15}\right)^{\mu_q} = \left(\frac{4p}{15}\right) = +1$, so non-residue factors occur with even total multiplicity, and the enumeration confirms that any two non-residue units (or one squared), together with the forced powers of 2, always reach the target. $\square$

Remark 3.3. The general composite-R engine (`theory.solvable_exact_general`), which evaluates the criterion of the composite meta-theorem directly in $(\mathbb{Z}/R)^*$, agrees with ground-truth divisor search on *every* composite admissible residual $R \in \{15, 27, 35, \ldots, 99\}$ (7,938 sampled hard primes per residual below 10^9, zero disagreements).

4 The sieve bound

We now prove the chain of Theorem 1.11, beginning with the base case, and then Lemma 1.9, its unconditional upgrade Theorem 1.10, the aggregate identities, and the density reduction.

Proof of Theorem 1.11, base case $\{3, 7\}$. Step 1 (exact criteria). Let p be a hard prime and set $n = (p+3)/4$, so that $(p+7)/4 = n+1$: *the two shifted forms are consecutive integers.* By Theorems 1.2 and 1.4, p is exceptional (both residuals fail) precisely when

$$n \text{ has no prime factor} \equiv 2 \pmod 3 \quad \text{and} \quad n+1 \text{ has no prime factor} \equiv 3, 5, 6 \pmod 7,$$

with $p = 4n - 3$ prime.

Step 2 (congruence bookkeeping). Fix one of the six hard classes $h \bmod 840$; then $n \equiv (h+3)/4 \pmod{210}$ is fixed, and the conditions at the primes $2, 3, 5, 7$ are determined by the class (in particular $3 \nmid n$ and $7 \nmid n+1$ automatically). It suffices to bound each class and sum.

Step 3 (containment in a sifted set). Let $z = x^{1/10}$. After discarding the $O(z)$ primes $p \le z$, every exceptional p gives $n \le (x+3)/4$, $n \equiv n_0 \pmod{210}$, such that for every prime q with $7 < q \le z$:

(a) $q \equiv 2 \pmod 3 \implies n \not\equiv 0 \pmod q$;

(b) $q \equiv 3, 5, 6 \pmod 7 \implies n \not\equiv -1 \pmod q$;

(c) $q \nmid 4n - 3$ (since $4n - 3 = p$ is prime, $p > z$).

Conditions (a), (b) only use "no forbidden factor up to z", which is implied by the full criteria — an upper bound needs no more.

Step 4 (sieve of dimension 2). This is a standard upper-bound sieve for the three linear forms n, $n+1$, $4n-3$, with sifting function

$$\omega(q) \;=\; \mathbf{1}_{q\equiv 2(3)} \;+\; \mathbf{1}_{q\,\mathrm{NQR}\,(7)} \;+\; 1 \qquad (q>7),$$

whose forbidden classes $0, -1, 3\cdot 4^{-1} \bmod q$ are pairwise distinct for $q \nmid 21$. By Dirichlet, ω has average $\kappa = \frac{1}{2}+\frac{1}{2}+1 = 2$ in the Halberstam–Richert sense $\sum_{q\le w}\omega(q)\log q/q = \kappa\log w + O(1)$. The Fundamental Lemma of sieve theory (see [7, Thm. 2.5] and [6, Ch. 6]) with $z = x^{1/10}$ bounds the exceptional set $E(x)$ of hard primes $p \le x$ failing both residuals:

$$\#E(x) \;\ll\; x \prod_{7<q\le z}\Big(1-\frac{\omega(q)}{q}\Big) \;\asymp\; x\,(\log z)^{-1}(\log z)^{-1/2}(\log z)^{-1/2} \;\asymp\; \frac{x}{(\log x)^2},$$

by Mertens' theorem and its arithmetic-progression form. Summing over the six classes preserves the bound. The higher exponents of the chain are deduced after Lemma 1.9 below. □

The verification behind Lemma 1.9 reduces to finitely many checks through the following elementary but load-bearing observation, which we isolate for scrutiny.

Lemma 4.1 (monotonicity reduction)**.** *For a class multiset C (classes v with multiplicities μ_v) let $D(C) = \{\sum_v e_v v : 0 \le e_v \le 2\mu_v\}$ denote the set of attainable divisor logs (before the p-shifts). If $\mu_v \le \mu'_v$ for all v, then $D(C) \subseteq D(C')$, where C' carries multiplicities μ'_v on the same classes; likewise enlarging the support enlarges D. Consequently a support S carries* some *failing configuration (any multiplicities ≥ 1 on S) if and only if the all-multiplicity-one configuration on S fails.*

Proof. Any admissible exponent vector for C is admissible for C', giving the inclusions. If a configuration on S with multiplicities $\mu_v \ge 1$ fails, the target lies outside $D(C)$, hence outside the subset $D(C_1)$ attained at multiplicities one — so C_1 fails too; the converse inclusion is trivial. The same argument applies verbatim after adjoining the p-shifts. □

Proof of Lemma 1.9. By Lemma 4.1 it suffices to check that for every support of size $(R-1)/2$ over the nonzero discrete logs and every class of $p \bmod R$, the target is reachable already at minimal multiplicities (exponent budget 2 per class, p-budget 2); success there kills every configuration on every superset support at any multiplicities. The check deliberately ignores the consistency relation $\prod(\text{classes}) \equiv 4^{-1}p$: it therefore examines a *superset* of the realizable configurations, and can only overstate the failing supports — the lemma is conservative. For $R = 19, 23$ a direct enumeration of all $(R-1)\binom{R-2}{(R-1)/2}$ (support, class of p) pairs suffices (437,580 resp. 7,759,752 pairs; code: `theory.verify_support_bound`). For the larger residuals the binomial count explodes, and the check is instead performed by a subset dynamic program (`theory.verify_support_bound_dp`): fold all supports into a table mapping each reachability mask to the maximal support size achieving it — valid because, by the same monotonicity, failure at any multiplicities implies failure at multiplicity one. The realized masks are heavily structured, so the state count stays tractable (3,001 states at $R = 31$ against 77.6 million supports; 19,530,971 states at $R = 107$, 331 seconds). All twelve residuals verify with zero violations; exact state counts are tabulated in Table 5. The bound is tight: the all-residue (Type I) configurations occupy exactly $(R-3)/2$ nonzero classes. □

Proof of Theorem 1.10. (i) Write $M = A_1 + \cdots + A_k$ with $A_i = \{0, v_i, 2v_i\}$, $k = |S|$, and let $H = \mathrm{Stab}(M) = \{h : M + h = M\}$, a subgroup of $\mathbb{Z}/d$. Kneser's addition theorem [9] (in the many-summand form obtained by iterating [16, Thm. 5.5]) gives

$$|M| \;\geq\; \sum_{i=1}^{k} |A_i + H| \;-\; (k-1)\,|H|.$$

Case $H = \{0\}$. Every $|A_i| = 3$, except that the (at most one) index i with $v_i = d/2$ has $|A_i| = 2$ (there $2v_i = 0$); so $|M| \geq (3k-1) - (k-1) = 2k$. Since $M \neq \mathbb{Z}/d$, $2k \leq d-1$, and d even forces $k \leq d/2 - 1$. *Case $|H| = h > 1$.* M is a union of H-cosets and $M \neq \mathbb{Z}/d$, so $|M| \leq d - h$. Split $k_0 = |S \cap H| \leq h - 1$ (distinct nonzero elements of H) and $k_1 = k - k_0$. For $v \in H$, $|A_v + H| = h$; for $v \notin H$, $A_v + H \supseteq H \cup (v + H)$, so $|A_v + H| \geq 2h$. Kneser's bound gives $d - h \geq h(k_0 + 2k_1 - k + 1) = h(k_1 + 1)$, so $k_1 \leq d/h - 2$ and $k \leq (h-1) + (d/h - 2) = h + d/h - 3$. For every divisor h with $2 \leq h \leq d/2$, the quadratic $h^2 - (\frac{d}{2} + 2)h + d \leq 0$ (its roots are 2 and $d/2$), i.e. $h + d/h - 3 \leq d/2 - 1$. Tightness: for S the nonzero even classes, $M(S)$ is the even subgroup and $|S| = d/2 - 1$.

(ii) In discrete-log coordinates on $(\mathbb{Z}/R)^* \cong \mathbb{Z}/d$, $d = R-1$, the divisor classes of m^2 contain the set $M(S)$ over the nonzero support S of $(p+R)/4$ at exponent budget 2 per class (Lemma 4.1: multiplicities ≥ 1 only enlarge budgets). If $|S| \geq d/2$, part (i) forces $M(S) = \mathbb{Z}/d$: every class — in particular the target $-m$ — is a divisor class of m^2, without invoking the p-powers, and (1.2) holds. Contrapositively, failure at R forces $|S| \leq d/2 - 1 = (R-3)/2$, which is the conclusion of Lemma 1.9 for arbitrary R.

(iii) The argument in (i) never used cyclicity. In a finite abelian G of even order g: in the H-trivial case each involution $v \in S$ has $|\{1, v, v^2\}| = 2$, so $|M| \geq (3k-t) - (k-1) = 2k - t + 1$ and $M \neq G$ gives $k \leq \lfloor (g+t-2)/2 \rfloor$; the $|H| = h > 1$ case is verbatim, giving $k \leq h + g/h - 3 \leq g/2 - 1$ over proper subgroups. For $G = (\mathbb{Z}/R)^*$, $t = 2^{\omega(R)} - 1$; with $\omega(R) \leq 2$, $t \leq 3$ and the bound is $\leq g/2$, so failure forbids at least $(g-1) - g/2 = \varphi(R)/2 - 1$ non-identity unit classes — dimension $\geq \frac{1}{2} - 1/\varphi(R)$. The general-$t$ clause is the same computation without the specialization: failure forbids at least $(g-1) - (g+t-2)/2 = (g-t)/2$ non-identity classes (minus one when the max is attained by $g/2 - 1$), and $2^{\omega(R)} \leq R^{O(1/\log\log R)} = R^{o(1)}$ gives the stated 0.44 threshold beyond an absolute bound. The upgrade to $\varphi(R)/2$ forbidden classes for $R \leq 107$ is the enumeration cited in the statement: the maximum support of a proper product-set is exactly $g/2 - 1$ for every composite admissible $R \leq 107$ (`theory.kneser_support_general`; e.g. $R = 95$: $g = 72$, maximum 35 over 583,455 realizable product-sets). That enumeration is a proof component of the $R \leq 107$ clause, on the same footing as the checks of Remark 1.7; the $\frac{1}{2} - 1/\varphi(R)$ tier needs Kneser alone. $\square$

Remark 4.2. Theorem 1.10(ii) is strictly stronger than what the sieve needs: at support $\geq (R-1)/2$ *every* class is reachable, for every target and every p. The strong form is also confirmed numerically (`theory.verify_support_bound_strong`): the maximum support of a non-full reachability mask equals $(R-3)/2$ exactly at every prime residual checked. Lemma 1.9 and its dynamic-program verifications now stand as independent machine confirmations of special cases (and remain the form carried by the Lean development).

Proof of Theorem 1.11, higher exponents. Write $n = (p+3)/4$; the relevant forms are n, $n+1$, $n+2 = (p+11)/4$, $n+3 = (p+15)/4$, $n+4 = (p+19)/4$, and $4n-3 = p$. Joint failure of $\{3, 7, 11\}$ implies, by Theorems 1.2, 1.4, 1.5: n free of primes $\equiv 2 \pmod 3$; $n+1$ free of non-residues mod

7; and, splitting by the two cases of Theorem 1.5: *branch (a)* — $n+2$ free of the five non-residue classes mod 11 (density $\frac{1}{2}$); *branch (b)* — $n+2$ free of the six classes $\{4,5,7,8,9,10\}$ mod 11 (density $\frac{3}{5}$). Each branch is a four-form sieve problem of dimension $1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}=\frac{5}{2}$, resp. $1+\frac{1}{2}+\frac{1}{2}+\frac{3}{5}=\frac{13}{5}$. Since $\frac{13}{5}>\frac{5}{2}$, the Type-II branch contributes $O(x/(\log x)^{13/5}) = o(x/(\log x)^{5/2})$: the bound is the maximum of the branch exponents, attained by the character branch (a). The Fundamental Lemma applied to each branch gives $O(x/(\log x)^{5/2})$ in total.

Adjoining $R=15$: by Theorem 1.6, failure at 15 is the single condition "$n+3$ free of the four classes $\{7,11,13,14\}$ mod 15" — one branch, density $\frac{4}{8}=\frac{1}{2}$ over the $\varphi(15)=8$ unit classes — raising each combined branch to dimension ≥ 3: joint failure of $\{3,7,11,15\}$ is $O(x/(\log x)^3)$.

For $\{3,7,11,15,19\}$, Theorem 1.10 (equivalently, at this residual, Lemma 1.9) splits failure at 19 into finitely many maximal-support branches, each forbidding an explicit set of at least 9 of the 18 unit classes mod 19 on $n+4$ — dimension $\geq \frac{1}{2}$. Crossing with the branches at 11 leaves finitely many combined branches, each a six-form problem of dimension $\geq \frac{7}{2}$ (those involving the Type-II case at 11 have dimension $\geq \frac{31}{10}+\frac{1}{2}>\frac{7}{2}$ and are again lower-order); the forbidden residues of the six forms are pairwise distinct for $q>19$, smaller primes being absorbed into the congruence class of n. Summing the finitely many branches gives $O(x/(\log x)^{7/2})$.

The general case is identical: for a finite admissible list $\mathcal{P}$ (prime, or composite with $\omega(R)\leq 2$), the forms $n+(R-3)/4$, $R\in\mathcal{P}$, are distinct integer shifts of n; Theorems 1.2, 1.4, 1.5, 1.6 handle $R=3,7,11,15$ and Theorem 1.10 handles every other residual (part (ii) for primes, part (iii) for composites), each contributing dimension δ_R through finitely many maximal-support branches — $\delta_R=\frac{1}{2}$ for primes and for composites ≤ 107, $\delta_R=\frac{1}{2}-1/\varphi(R)$ for larger composites (the two tiers of part (iii)) — plus 1 for primality: total dimension $\geq 1+\kappa(\mathcal{P})$, with forbidden residues pairwise distinct for $q>\max\mathcal{P}$ and smaller primes absorbed into the congruence class of n. For the fifteen primes up to 107 this gives exponent $\frac{17}{2}$; for the full 27-element admissible list, $\frac{29}{2}$. □

Proof of Proposition 1.12. $k=ap^2$ divides $m^2=p^2a^2$ and $k\equiv -m$ if and only if $ap^2\equiv -pa$, i.e. $p\equiv -1 \pmod R$; $k=a^2p$ divides m^2 and $a^2p\equiv -pa$ if and only if $a\equiv -1$, i.e. $p\equiv -4 \pmod R$ by (1.3). Integrality of b,c is automatic. Scanning all shapes $k=a^jp^i$ ($j,i\leq 2$) yields no further p-only condition. The failure of both families states that the two forms $(p+1)/2$ and $p+4$ (in n: $2n-1$ and $4n+1$) are free of primes $\equiv 3 \pmod 4$ — two further sifting conditions of dimension $\frac{1}{2}$ each on forms distinct from all previous ones. □

Empirically the two families alone cover 74% of hard primes below 10^9 — and none of the four primes of Section 6 that admit a unique working residual ≤ 107.

Remark 4.3. By Theorem 1.10, every residual — prime or composite — adds $\frac{1}{2}$ to the exponent, so the chain's exponent is bounded only by the number of residuals used; the constants $O_{\mathcal{P}}$, however, grow with $|\mathcal{P}|$ (branch counts and the remainder term of the Fundamental Lemma). The chain is a concrete instantiation of Theorem 1.13 with explicit constants. One caveat frames its scope, and it is fundamental: no fixed finite list can be pushed to *emptiness* by upper-bound sieving: the count bound $x/(\log x)^{\kappa}$ diverges for every fixed κ, and extrapolating the calibration of Section 6 to any fixed list yields a positive density constant, i.e. infinitely many expected exceptional primes with the record minimal residual growing slowly without bound. The correctly posed remaining problem is a slowly growing bound $R_{\min}(p)\ll f(p)$, which requires existence (lower-bound) technology beyond upper-bound sieves.

Proof of Theorem 1.13. Fix $A > 0$. Choose $B = B(A)$ so large that

$$\sum_{\substack{3 \le R \le B \\ R \equiv 3\ (4)}} \frac{1}{\varphi(R)} \ \ge \ \sum_{\substack{3 \le R \le B \\ R \equiv 3\ (4)}} \frac{1}{R} \ > \ A; \tag{4.1}$$

this is possible since the reciprocal sum over the progression $R \equiv 3 \pmod 4$ diverges, and $\varphi(R) \le R$. Let $S(A) = \{R \equiv 3 \pmod 4 : 3 \le R \le B\}$ and $M = 840 \cdot \prod_{R \in S(A)} R$.

Write $n = (p+3)/4$ as before, so $(p+R)/4 = n + (R-3)/4$ for each $R \in S(A)$: a family of $|S(A)|$ integer linear forms in n with pairwise distinct shifts, together with the form $4n - 3 = p$. Set $z = x^{1/10}$ and discard the $O(z)$ primes $p \le z$. Fix a residue class $n \equiv n_0 \pmod M$ compatible with p hard; there are finitely many, and it suffices to bound each. Within such a class, for every $R \in S(A)$ the quantities $t \equiv -4^{-1}p^2$ and $p^{-1} \bmod R$ are constant, so the class set $S_R(p) = \{t, tp^{-1}, tp^{-2}\} \bmod R$ of Proposition 2.2 is a fixed nonempty set of $t_R = t_R(n_0) \in \{1,2,3\}$ classes. By the contrapositive of Proposition 2.2, if every residual in $S(A)$ fails at p, then for each $R \in S(A)$ the form $n + (R-3)/4$ has *no* prime factor in $S_R(p)$; and $4n - 3$, being prime and greater than z, has no prime factor $\le z$.

Sift the class $\{n \le (x+3)/4 : n \equiv n_0\ (M)\}$ by the primes $q \le z$, $q \nmid M$. A prime q forbids: the class $n \equiv 3 \cdot 4^{-1} \pmod q$ (from the primality of $4n-3$), and, for each $R \in S(A)$ with $q \bmod R \in S_R(p)$, the single class $n \equiv -\frac{R-3}{4} \pmod q$ (a prime factor q of the form $n + (R-3)/4$ in a guaranteed-success class cannot occur). Thus

$$\omega(q) \ = \ 1 \ + \ \#\{R \in S(A) : q \bmod R \in S_R(p)\},$$

which is bounded by $|S(A)| + 1$; the forbidden residues of distinct forms are pairwise distinct for $q > B$, since the shifts $(R-3)/4$ differ by less than B and $q \nmid M$. For each R, the density of primes q with $q \bmod R \in S_R(p)$ is $t_R/\varphi(R)$ by Dirichlet's theorem, so ω has average

$$\kappa \ = \ 1 + \sum_{R \in S(A)} \frac{t_R}{\varphi(R)} \ \ge \ 1 + \sum_{R \in S(A)} \frac{1}{\varphi(R)} \ > \ 1 + A,$$

in the Halberstam–Richert sense, by (4.1). The Fundamental Lemma [7, Thm. 2.5] in dimension κ then bounds the sifted count by $O_A\Big(x \prod_{q \le z}(1 - \omega(q)/q)\Big) = O_A(x(\log x)^{-\kappa}) = O_A\Big(x/(\log x)^{1+A}\Big)$, and summing over the finitely many classes $n_0 \bmod M$ preserves the bound. $\square$

Remark 4.4. Theorem 1.13 is the strongest statement the pure sieve can give: density $O((\log x)^{-A})$ for every A, but not emptiness. The conjecture for hard primes would follow from a *finite covering hypothesis*: some finite S admits, for every hard prime, a residual $R \in S$ with (1.2) (the converse is not automatic — the conjecture bounds $R_{\min}(p)$ by $2p$, not uniformly). The data to 10^{11} is covered by $S = \{3, 7, 11, \dots, 107\}$ — yet both the calibration (Remark 4.3) and the conditional construction of Section 7 predict that any fixed list eventually fails.

4.1 The uniform chain

Theorem 1.13 fixes the list $S(A)$ once A is fixed; letting the list *grow with* x is purely a question of uniformity in the constants of Theorem 1.11 — the branch count per residual, the class modulus, and the sieve constants in growing dimension. Auditing those constants shows the growth is affordable for lists of length a small multiple of $\log\log x$, which yields the strongest unconditional statement of this paper.

Theorem 4.5 (uniform chain)**.** *There are absolute constants $\varepsilon, c > 0$ and an (ineffective) x_0 such that for all $x \geq x_0$,*

$$\#\{\textit{hard primes } p \leq x : R_{\min}(p) > \varepsilon \log\log x\} \;\leq\; x\,\exp\!\big(-c\,(\log\log x)^2\big).$$

In particular, all but $O(xe^{-c(\log\log x)^2})$ hard primes $p \leq x$ possess an explicit, divisor-exhibited residual certificate at some admissible $R \leq \varepsilon \log\log x$ — unconditionally.

Proof. Write $L = \log\log x$, $B = \varepsilon L$, and let $\mathcal{R}$ be the admissible residuals $3 \leq R \leq B$ ($|\mathcal{R}| \sim B/4$). Discard the $O(\sqrt{x})$ primes $p \leq \sqrt{x}$. A hard prime with $R_{\min}(p) > B$ fails every $R \in \mathcal{R}$; we bound the count of such p by a single sieve application per *branch*, as follows.

Dimensions. Each $R \in \mathcal{R}$ contributes a sifting dimension δ_R: the exact criteria at $R = 3, 7, 11, 15$ and Theorem 1.10(ii) at primes give $\delta_R = \frac{1}{2}$; composites ≤ 107 give $\frac{1}{2}$ by the sharpened tier of Theorem 1.10(iii); and every remaining composite — with *no* restriction on $\omega(R)$ — gives $\delta_R \geq \frac{1}{2} - (2^{\omega(R)-1} + 1)/\varphi(R) \geq 0.44$ beyond an absolute bound R_1, by the general-t clause of Theorem 1.10(iii). Hence $\kappa := 1 + \sum_{R\in\mathcal{R}} \delta_R \geq 1 + 0.44(|\mathcal{R}| - R_1) \geq 1 + B/10$ for B large.

Branches. For each R, failure confines the support of $(p+R)/4$ to one of at most $2^{\varphi(R)}$ failing supports S_R (crudely, all subsets); fixing $(S_R)_{R\in\mathcal{R}}$ and the class n_0 of $n = (p+3)/4$ modulo $\widetilde{M} = \operatorname{lcm}(840, \prod_{\ell\leq B} \ell) = e^{O(B)}$ costs a factor $\prod_R 2^{\varphi(R)} \cdot e^{O(B)} = e^{O(B^2)}$ branches.

Sieve. Within a branch, sift the integers $n \leq (x+3)/4$, $n \equiv n_0 \pmod{\widetilde{M}}$, by the primes $\ell \in (B, z]$, $z = x^{1/s}$: the forbidden classes mod ℓ are $n \equiv 3 \cdot 4^{-1}$ (so that $p = 4n - 3$ keeps no factor $\leq z$) and, for each R with ℓ mod R outside S_R, $n \equiv -(R-3)/4$ — pairwise distinct for $\ell > B$, so $\omega(\ell) \leq |\mathcal{R}|+1 < \ell/2$. The density condition holds with the dimension κ above and (Ω)-constant $O(B \log B)$, by Mertens' theorem in the progressions modulo $R \leq B$ (Siegel–Walfisz range, whence the ineffective x_0). Apply Brun's sieve of order $r \asymp \kappa$ [7, Ch. 2] at $s \asymp \kappa$: since the sifted sequence consists of *integers* in a progression — no primality is detected anywhere — the remainder is $\sum_{d\leq z^r} (B+2)^{\omega(d)} \ll x^{1/10} e^{O(BL)} = x^{1/10+o(1)}$ for a suitable absolute choice of the implied constants, and the main term is

$$\ll \frac{x}{\widetilde{M}} \prod_{B<\ell\leq z} \Big(1 - \frac{\omega(\ell)}{\ell}\Big) \;\leq\; x \exp(-\kappa(L - O(\log B)) + O(B\log B)).$$

Accounting. Multiplying by the branch count,

$$\begin{aligned}\#\{p \leq x : R_{\min}(p) > B\} \;&\leq\; e^{O(B^2)}\, x\, \exp\big(-\kappa L + O(B\log B) + O(\kappa \log B)\big)\\ &\leq\; x\, \exp(-BL/10 + O(B^2)),\end{aligned}$$

and with $B = \varepsilon L$ the exponent is $-\varepsilon L^2/10 + O(\varepsilon^2 L^2) \leq -\varepsilon L^2/20$ for ε below an absolute threshold. ∎

Remark 4.6 (the ceiling, and what lies beyond). Theorem 4.5 sits at the exact boundary of this method family: the branch factor $e^{\Theta(B^2)}$ against the sieve gain $e^{-BL/10}$ forces $B \ll L$, and independently the class modulus $\widetilde{M} = e^{O(B)}$ and the Mertens-in-progressions inputs leave their valid ranges at $B \gg L$ — three walls detonating at the same scale, so $\exp(-cL^2)$ is the honest unconditional frontier here even under substantial improvements to the branch count. Vaughan's bound $x\exp(-c(\log x)^{2/3})$ [17] remains far stronger as a pure density statement; what Theorem 4.5 adds is structural: an explicit certificate at a residual of size $O(\log\log x)$ for almost every hard prime, a statement mean-value methods do not address. Passing the $\exp(-cL^2)$ ceiling toward Vaughan and beyond is exactly what the measured hypotheses of the ladder program (Section 5) would buy: the calibrations $J = (\log x)^\theta$ of Theorem 5.9 are out of reach of every unconditional variant considered here.

5 The Burgess–reciprocity ladder

Theorem 1.10 closed the fixed-list axis, and Theorem 4.5 pushed the growing-list axis to its unconditional ceiling $\exp(-c(\log\log x)^2)$ (Remark 4.6); no list-based bound can reach emptiness (Remark 4.3). This section develops the program that opens past the ceiling: a *p-adapted* family of residuals, chosen through quadratic reciprocity, along which the failure modes are localized, partially proved, and reduced to a single measured hypothesis. Entry points and archives for all measurements cited here are in Appendix A.

5.1 Necessity, purity, and the selected residual

Lemma 5.1 (composite reciprocity)**.** *Let $R \equiv 3 \pmod 4$ be any admissible residual and q an odd prime with $q \mid (p+R)/4$ and $\gcd(q,R) = 1$. Then the Jacobi symbol satisfies $\left(\frac{q}{R}\right) = \left(\frac{p}{q}\right)$.*

Proof. $q \mid p+R$ gives $R \equiv -p \pmod q$; Jacobi reciprocity with $R \equiv 3 \pmod 4$ gives $\left(\frac{q}{R}\right) = (-1)^{\frac{q-1}{2}}\left(\frac{R}{q}\right) = (-1)^{\frac{q-1}{2}}\left(\frac{-1}{q}\right)\left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)$ — the proof of Theorem 1.8 verbatim, which never used primality of R. $\square$

Lemma 5.2 (Jacobi necessity)**.** *For every admissible R: if no prime factor of $a = (p+R)/4$ has Jacobi symbol $\left(\frac{\cdot}{R}\right) = -1$, then* (1.2) *fails at R.*

Proof. All factors of a Jacobi residues gives $\left(\frac{a}{R}\right) = +1$; by consistency (1.3), $\left(\frac{p}{R}\right) = \left(\frac{4a}{R}\right) = +1$ as well, so every prime factor of $m = pa$ has symbol $+1$, hence every divisor $k \mid m^2$ has $\left(\frac{k}{R}\right) = +1$, while the target has $\left(\frac{-m}{R}\right) = \left(\frac{-1}{R}\right) = -1$. $\square$

Call R *pure* (at hard primes) if the converse also holds: a Jacobi non-residue factor present implies success. Theorems 1.2, 1.4 and 1.6 say exactly that $R = 3, 7, 15$ are pure; the census below shows they are the *only* pure residuals up to 107, with the conditional failure rate at the others ranging from 1.3% ($R = 23$) to 73% ($R = 67$).

Now run Lemma 5.1 backwards. For a hard prime p let $q = q(p)$ be its least Legendre non-residue (hard primes are residues at $3, 5, 7$ by Corollary 2.3, so $q \geq 11$; Burgess bounds $q \ll p^{1/(4\sqrt{e})+\varepsilon}$ unconditionally [2] and GRH gives $q \ll (\log p)^2$ [1]; the observed maximum below 10^{11} is 83), and let the *selected residual* be $R_0 = (-p) \bmod 4q$, so that $R_0 \equiv 3 \pmod 4$ automatically and $q \mid (p+R_0)/4$. By Lemma 5.1, $\left(\frac{q}{R_0}\right) = \left(\frac{p}{q}\right) = -1$: the all-residue failure mode of Lemma 5.2 is impossible at R_0 *by construction.* The *q-ladder* is the progression $R_j = R_0 + 4qj$, along which the same holds at every rung.

5.2 The census

Computed over every one of the 1,587,581 hard primes below 10^9, with mask-free window samples of 10,000 primes per half-decade beyond, and the complete 10^{11} deep tail (Appendix A), the census is summarized in Table 2.

Failure anatomy (all 82,808 first-rung failures below 10^9): 97.5% are *true budget* failures — the subgroup generated by the factor classes contains the target and only the bounded exponents miss it — and 29% miss exactly one unit class, the target itself. No failure has more than 5 distinct non-identity factor classes (Theorem 1.10 would allow $(R-3)/2$), with the conditional failure probability decaying by a factor ≈ 7 per additional class; and the Jacobi non-residue mass sits at the parity minimum forced by consistency in 97.9% of failures — the mechanism of Theorem 1.6's proof, observed at every residual.

5.3 What is proved

Theorem 5.3 (ladder chain)**.** *Fix a prime $q \geq 11$ and $J \geq 1$. The number of hard primes $p \leq x$ with $\left(\frac{p}{q}\right) = -1$ failing every rung R_j, $0 \leq j < J$, of their q-ladder is $O_{q,J}(x/(\log x)^{1+\kappa_J})$, where $\kappa_J = \sum \delta_{R_j}$ over the rungs with at most two prime factors, with $\delta_R = \frac{1}{2}$ for R prime or composite ≤ 107 and $\delta_R = \frac{1}{2} - 1/\varphi(R)$ for larger composites; in particular $\kappa_J \geq J'/2 - \sum_j 1/\varphi(R_j)$ with J' the number of such rungs.*

Proof. The shifted forms $(p + R_j)/4 = n + (R_0 - 3)/4 + qj$ are distinct integer shifts of $n = (p+3)/4$, and Theorem 1.10 — part (ii) at prime rungs, part (iii) at composite rungs with $\omega \leq 2$ — makes failure at each usable rung a sifting condition of dimension $\geq \delta_{R_j}$ (the two tiers of part (iii) at composite rungs), exactly as in the proof of Theorem 1.11, which applies verbatim to this p-adapted list. $\square$

Theorem 5.4 (first rung)**.** *Fix a prime $q \geq 11$. Among hard primes $p \leq x$ with least non-residue $q(p) = q$, the proportion failing the selected residual R_0 is $\ll_q (\log x)^{-\eta_q} \to 0$, where $\eta_q = \min 1/\varphi(R_0) > 1/(4q)$, the minimum over the finitely many selected residuals $R_0 < 4q$ that arise for q.*

Proof. The condition $q(p) = q$ is a union of residue classes modulo $M = 840 \prod_{11 \leq \ell \leq q} \ell$; fix one, which fixes R_0. By Proposition 2.2, any prime factor u of $(p + R_0)/4$ in the universal class $-4^{-1} \bmod R_0$ certifies success via $k = up^2$, so a failing p has $(p + R_0)/4$ free of an entire congruence class of primes: a sifting condition of dimension $1 + 1/\varphi(R_0)$ (with primality), and the Fundamental Lemma bounds the failing count by $\ll x/(\log x)^{1+1/\varphi(R_0)}$ against class size $\asymp x/\log x$ (Siegel–Walfisz). Summing over the classes, the proportion is governed by the worst class, giving η_q; and $\varphi(R_0) < R_0 < 4q$ gives $\eta_q > 1/(4q)$. The bound is uniform for $q \ll \log\log x$. $\square$

Theorem 5.5 (proxy ladder, two-sided)**.** *Fix q, a class (fixing $R_0, R_1, \dots$), and J with $\kappa = \sum_{j \leq J} 1/\varphi(R_j)$ below an absolute constant. Let $\tilde{E}_J(x)$ be the set of hard primes in the class whose shifted values $(p + R_j)/4$, $0 \leq j \leq J$, all avoid prime factors in their universal classes (so $E_j \subseteq \tilde{E}_j$ for every $j \leq J$). Then*

$$x/(\log x)^{1+\kappa} \ll_{q,J} \#\tilde{E}_J(x) \ll_{q,J} x/(\log x)^{1+\kappa} :$$

the avoidance ladder decays by the factor $(\log x)^{-1/\varphi(R_{j+1})}$ at every rung.

Proof sketch. The upper bound is the Fundamental Lemma as in Theorem 5.4. The lower bound is a beta-sieve in dimension $\kappa < 1$ on the prime-supported sequence, with level $x^{1/2-\varepsilon}$ from Bombieri–Vinogradov; integers with a forbidden prime factor above the sifting range are removed by Brun–Titchmarsh upper bounds in the progressions they define, whose total is absorbable for κ small. □

Hypothesis 5.6 (P; proportionality)**.** *For each fixed q, class, and J there are $c_P > 0$ and x_0, with c_P bounded below uniformly in the rung $j \le J$ (uniformity in q is part of the hypothesis), such that $\#E_j(x) \ge c_P\,\#\tilde{E}_j(x)$ for all $j \le J$ and $x \ge x_0$, where E_j is the set of true ladder failures through rung j: failures are a positive proportion of their necessary-condition set.*

Measured: $c_P \approx 0.097$–0.12 at rung 0 and 0.13–0.16 at rung 1, flat across two decades to 10^{11} (Appendix A). Under Hypothesis 5.6, since $E_{j+1} \subseteq \tilde{E}_{j+1}$,

$$\#E_{j+1} \;\le\; \#\tilde{E}_{j+1} \;\ll\; (\log x)^{-1/\varphi(R_{j+1})}\,\#\tilde{E}_j \;\le\; (\log x)^{-1/\varphi(R_{j+1})}\,c_P^{-1}\,\#E_j,$$

so the ladder failure count decays at every *fixed* rung, for $x \ge x_0(q, j, \delta)$ — the per-rung decay Hypothesis 5.8 below posits, obtained from Hypothesis 5.6 rung by rung, though not with the uniformity in j and q that Hypothesis 5.8 demands.

Theorem 5.7 (half-dimensional failure lower bound)**.** *Call a class c mod M (M as in the proof of Theorem 5.4) admissible if its R_0 is composite with a prime factor $r_1 \equiv 3 \pmod 4$ such that $\left(\frac{q}{r_1}\right) = +1$ (and $\left(\frac{2}{r_1}\right) = +1$ when $2 \mid a$). On every admissible class,*

$$\begin{aligned}\#E_0(x) \;&\ge\; \#\{p \le x : p \equiv c \pmod M,\ \textit{every prime factor of } (p+R_0)/4 \textit{ a QR mod } r_1\}\\ &\gg\; \frac{x}{(\log x)^{3/2}}.\end{aligned}$$

Proof sketch. Membership forces failure by Proposition 2.1 at r_1: all factors of a are residues mod r_1 by hypothesis, and $\left(\frac{p}{r_1}\right) = +1$ follows *for free* from consistency $a \equiv 4^{-1}p$, so every divisor of m^2 is a residue while $\left(\frac{-m}{r_1}\right) = \left(\frac{-1}{r_1}\right) = -1$. The membership condition sifts the $(r_1-1)/2$ non-residue classes mod r_1 — dimension exactly $\frac{1}{2}$ — and Iwaniec's half-dimensional sieve [8] produces the matching-order lower bound for the full condition: at dimension $\frac{1}{2}$ the sifting range reaches the Bombieri–Vinogradov level, leaving at most one large non-residue factor, removed by a switching argument. □

Theorem 5.7 is the program's first unconditional *lower* bound on true ladder failures (6.6% of first rungs are admissible; every all-residue member found in the data fails as required, e.g. $p = 5{,}544{,}361$ with $R_0 = 51$, $r_1 = 3$, $a = 31 \cdot 61 \cdot 733$). Against Hypothesis 5.6 it gives the two-sided bracket $(\log x)^{-(1/2-1/\varphi(R_0))} \ll \#E_0/\#\tilde{E}_0 \le 1$: what remains of the hypothesis is exactly the closing of this log-gap, the parity obstruction in its sharpest local form.

5.4 The conditional theorem, and honest scope

Hypothesis 5.8 (B; per-rung decay)**.** *Fix a rung-count function $J(x) \to \infty$ (as calibrated after Theorem 5.9). There are $\delta > 0$ and x_0 such that for $x \ge x_0$, all $q \le (\log x)^2$ and all $0 \le j < J(x)$: $\#E_{j+1}(x) \le (1-\delta)\,\#E_j(x) + O(\sqrt{x})$.*

The $j = 0$ case is Theorem 5.4; every fixed rung follows from Hypothesis 5.6 (non-uniformly in j and q); the measured survival factors $1-\delta$ are 0.05–0.14 per rung — 86–95% of the surviving failures resolve at each rung — with no drift in p.

Theorem 5.9 (conditional almost-all bound)**.** *Assume Hypothesis 5.8. Then all but*

$$O\Big(x\, e^{-\delta J(x)} \;+\; x\, 2^{-\pi((\log x)^2)}\Big)$$

hard primes $p \le x$ *satisfy* $R_{\min}(p) \le 4\, q(p)\, (J(x)+1) \le (\log x)^{2+o(1)} J(x)$*.*

Proof. Hard primes with $q(p) > (\log x)^2$ are residues at every prime in $(7, (\log x)^2]$, a half-class condition per modulus, and the large sieve bounds their count by the second term. For the rest, iterate Hypothesis 5.8 $J(x)$ times; a survivor is the only way to lack a certificate on the ladder. □

With $J(x) = (\log\log x)^2$ — a mild extrapolation of the nine measured rungs — the exceptional set is $\ll x \exp(-\delta(\log\log x)^2)$: this calibration is now delivered *unconditionally*, and with the stronger bound $R_{\min} \le \varepsilon \log\log x$, by Theorem 4.5, so Hypothesis 5.8's real content begins past the ceiling of Remark 4.6: with $J(x) = (\log x)^{\theta}$, $\theta > 2/3$, Theorem 5.9 would pass Vaughan's bound. The hypothesis-strength required scales with the ambition, and each increment is testable before it is assumed.

Remark 5.10 (scope)*.* Theorem 5.9 is an almost-all statement under a measured hypothesis; three things it is not. It is not unconditional: closing Hypothesis 5.6's log-gap is a parity problem, and Theorem 5.7 sits at dimension $\frac{1}{2}$ precisely because that is the unique historically known opening in the parity wall. It is not a density record: Vaughan's bound remains asymptotically stronger unconditionally. And it is not a path to the full conjecture by itself: emptiness is beyond upper-bound methods (Remark 4.3), and excluding the individual conspiratorial prime — a super-record dodging $J(p)$ rungs, of which the record prime of Section 6 is an empirical foreshadowing — is a simultaneous-multiplicative-structure statement of prime-tuple difficulty. What the ladder does is localize the conjecture for hard primes into one measured, twice-bracketed counting inequality, with every sub-statement current tools can decide now decided.

Remark 5.11 (why the walls are real: two obstruction computations)*.* Two computations pin the walls down. (i) *Budget failure is not a character event.* In the model at $R = 11$ (generator 2), the configurations with class set $\{2, 7\}$ and multiplicities $(1, 1)$ versus $(4, 2)$ have the *same* class set and the same product $a \equiv 3 \pmod{11}$ — hence identical values under every Dirichlet character mod 11 — yet the first fails the target 8 ($\equiv -m$ for the consistent $p \equiv 1 \pmod{11}$) and the second attains it. No character-based machinery (pretentious dichotomies included) can separate the two, and 97.5% of measured first-rung ladder failures are of this budget type. (ii) *Weighted detection is equivalent to Hypothesis 5.6.* The certificate detector is already maximally relaxed — by Proposition 2.2, any *divisor* of the shifted value in the universal class certifies, so almost-prime weakenings of the detected object change nothing — and the first moment of any Bonferroni/Selberg weighting over E_j is itself a failure count: carrying the scheme through is equivalent, up to constants, to Hypothesis 5.6 itself. The parity gap of Theorem 5.7 is therefore intrinsic to every relaxation of this type, not an artifact of the chosen detector.

6 Computations to 10^{11}

All code and data are in the accompanying repository. Certificates are found by direct integer factorization of a (trial division; the factorization of $m^2 = p^2 a^2$ follows) and scanning divisors for the class condition (1.2); every certificate is re-verified in exact integer arithmetic, $4abc = p(bc + ac + ab)$.

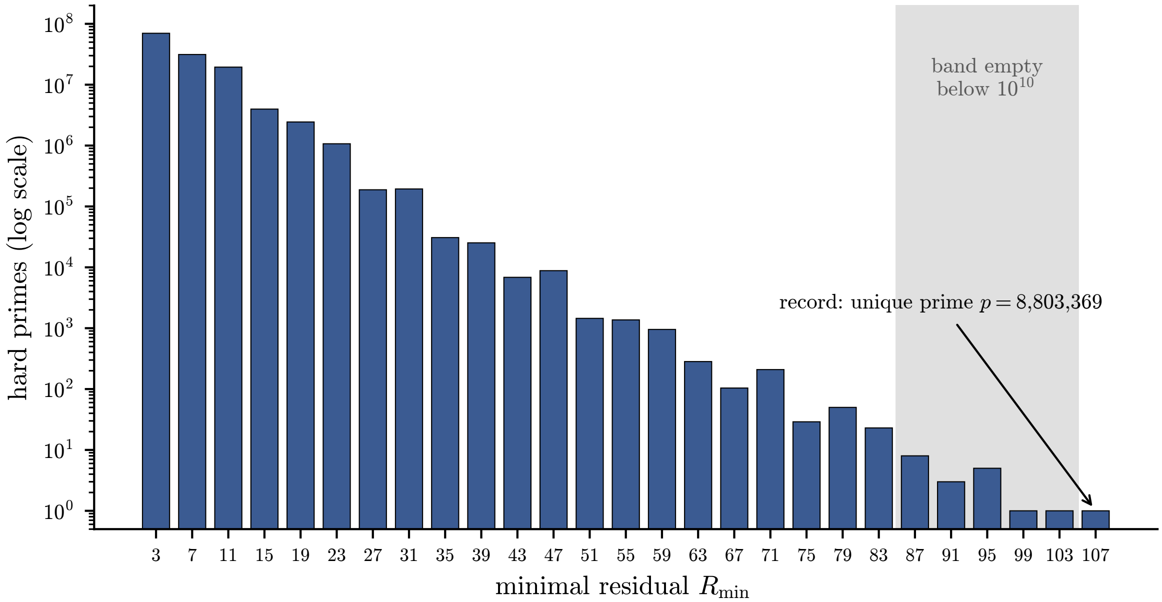

Figure 1: The data of Table 3 on a logarithmic count scale. The head is dominated by $R = 3, 7, 11$; the tail thins geometrically; the shaded band 87–103 was empty below 10^{10} and its late population matched the calibrated prediction (see the confirmed-prediction bullet below); the record $R = 107$ is attained by a single prime below 10^7.

6.1 Data

Every one of the 128,671,219 hard primes $p < 10^{11}$ possesses a residual certificate with $R \leq 107$. The verification behind this claim is graded by dataset: exhaustive reconstruction and exact re-checking of every entry below 10^9; systematic samples (143,000 resp. 257,343 entries) at 10^{10} and 10^{11}, together with the *complete* tails (842 entries with $R \geq 51$ resp. 20,151 with $R \geq 43$) re-checked with full minimality — Table 6 gives the per-dataset detail. The minimal residuals distribute as in Table 3 and Figure 1.

Three features of the distribution deserve comment.

- $R_{\max} = 107$, attained by the single prime $8{,}803{,}369 < 10^7$ and never exceeded through 10^{11}. In the language of Corollary 2.3: its 27 shifted values contain only 43 distinct odd primes (typical: 45–52), and 81% of them satisfy $\left(\frac{p}{q}\right) = +1$ (typical: 44–60%) — a $\approx 4\sigma$ fluctuation ($z = 4.1$ against the binomial null of independent unit signs) in the statistic that Theorem 1.8 identifies as governing joint failure.

- **A confirmed prediction.** The *calibrated independence model* treats the per-residual failure events as independent across R, with per-residual failure rates fitted from the complete 10^9 solvability masks and a single deep-tail correlation factor calibrated on the $R > 83$ counts; it is a heuristic, not a theorem, and its role is to make falsifiable predictions. Below 10^{10} no prime had minimal $R \in \{87, 91, 95, 99, 103\}$; the model attributed this gap to small-number statistics (expected counts ≤ 0.12 per value at 10^9) and predicted that deep-tail primes (failing every $R \leq 83$) land at $87, 91, 95, 99, 103, \geq 107$ with probabilities $0.53, 0.11, 0.22, 0.03, 0.05, 0.06$ (the last being the total mass at or beyond

107). Between 10^{10} and 10^{11} the five values acquired $8, 3, 5, 1, 1$ primes respectively (all in $(1.3 \times 10^{10}, 10^{11})$), and none passed 103: the observed conditional distribution over the 19 deep-tail primes $(8, 3, 5, 1, 1, 1$ including the record) matches the prediction.

- Among all 1,587,581 hard primes below 10^9, exactly *four* admit a unique working residual ≤ 107 (8,803,369 with 107; 142,361,209 with 59; 287,567,281 with 83; 794,037,841 with 63) — the entire obstruction to a shorter covering list. A greedy cover of the full solvability table uses 18 residuals; a packing argument gives a lower bound of 12.

6.2 Calibration of the base case of Theorem 1.11

The proportion of hard primes failing both $R = 3$ and $R = 7$, multiplied by $\log x$, is constant to within 1% across three decades of the 10^9 data (Table 4).

That is, the joint exceptional density is $\approx 5.17/\log x$: the decay matches the order predicted by the base case of Theorem 1.11, and 5.17 estimates the (conjectural) Selberg–Delange constant. The exceptional set is density zero but decays slowly: 26.5% of hard primes below 10^9 fail both residuals; all are settled by deeper residuals of the list. The fitted per-residual decay exponents are consistent with the sieve prediction $\kappa = \frac{1}{2}$ where exact criteria exist: $\hat{\kappa} = 0.539$ for $R = 7$, and $\hat{\kappa} = 0.387$ for $R = 3$, where the short fitting range and Selberg–Delange lower-order terms bias the exponent downward.

6.3 Failure taxonomy

Sampling the complete solvability table at 10^9: failures at $R = 7$ are 100% Type I (as Theorem 1.4 requires); at $R = 11$, 89.7% Type I and 10.3% the budget pattern of Theorem 1.5(b); the Type-I share broadly decreases with $\varphi(R)$ (e.g. 51% at $R = 103$, though not monotonically), as hitting one specific class among $\varphi(R)$ grows harder while the character dichotomy stays binary. Taken together, the data of this section matches the framework of Sections 2–4: each distributional feature — the head, the geometric tail, the once-empty band, the static record — has a structural or heuristic explanation (the exact criteria for the head; the calibrated model, built on the sieve dimensions, for the tail and the band; Theorem 1.8 for the record).

7 Open problems

1. **Finite covering.** Prove that some finite admissible set S covers all hard primes (Theorem 1.13 gives density $(\log x)^{-A}$ for every A; emptiness is beyond the sieve).

2. **Beyond fixed lists.** The support bound is proved unconditionally for every residual, prime and composite (Theorem 1.10), and Theorem 4.5 shows the two walls of list growth — per-branch forbidden classes depending on $p \bmod R$, and the growing branch count — are affordable up to lists of length $\varepsilon \log \log x$, but no further (Remark 4.6); the fixed-list format itself cannot reach emptiness (Remark 4.3). Passing the ceiling by structure rather than hypothesis would need a poly(R) *branch classification*: every failing configuration contained in polynomially many kernel or coset-progression branches — a Kneser/Freiman-type structure problem for bounded-exponent product sets, whose $\omega(R) \leq 2$ extremal cases are exactly Theorem 1.10(i)/(iii) and the `kneser_support_general` data. The proof of Theorem 1.10 shows the extremal failing supports are *subgroup-trapped* (the stabilizer H is nontrivial and S concentrates on H up to at most $d/|H| - 2$ outliers); exploiting this

structure *uniformly in R* — fewer, p-independent branches — is now a well-posed additive-combinatorics question, and the structurally right target remains a slowly growing bound on $R_{\min}(p)$.

3. **The constant 5.17.** The joint $\{3, 7\}$-exceptional density is measured as $\approx 5.17/\log x$, stable to 1% across three decades. Conjecturally 5.17 is a Selberg–Delange-type constant: the count should satisfy $E(x) \sim C\,x/(\log x)^2$ with $C = c_0 \prod_q \delta_q$, the local factors δ_q recording, for each prime q, the probability that q divides neither form in a forbidden class. Proving the asymptotic (not just the upper bound) requires a two-form Selberg–Delange or shifted-convolution analysis — open, and a natural test problem for such methods (see also Remark 1.14 for the density context).

4. **Structural reformulations and complexity.** Theorem 1.8 invites a graph formulation: on the bipartite graph joining primes q to residuals R by $q \mid (p+R)/4$, the edge characters $\left(\frac{q}{R}\right)$ collapse to vertex signs $\left(\frac{p}{q}\right)$, and joint Type-I failure is an all-plus vertex assignment — a percolation-style object whose structure could organize the growth law. Separately, the meta-theorem's enumeration complexity is unresolved: the DP state count grows empirically by $\approx 2.8\times$ per successive prime residual (271 at $R = 19$ to 1.95×10^7 at $R = 107$); is the equivalence-state complexity genuinely exponential in R? (For the support-bound property itself the question is settled: Theorem 1.10 is a constant-size certificate. The open version concerns the full exact criteria.)

5. **The record and the covering hypothesis.** Theorem 1.8 explains the record prime quantitatively (see Corollary 2.3) and yields a conditional resolution of the covering question: under Dickson's prime-tuple conjecture, for every B there are infinitely many hard primes failing every residual $\leq B$ — fix the forced smooth parts of the shifted values (character-neutral by Corollary 2.3(i)), make the cofactors prime in prescribed residue classes, and apply Proposition 2.1. So no fixed finite list suffices, conditionally, and the record grows without bound (at the pure-Type-I level, like $\log x/\log\log x$). Unconditional versions appear out of reach: already "infinitely many hard p fail $R = 3$" contains a two-form prime-tuple statement. What remains open is an unconditional upper bound on $R_{\min}(p)$ — and this is not an incremental item: by Proposition 1.1, $R_{\min}(p) \leq 2p$ is *equivalent* to the conjecture at p, so an unconditional bound $R_{\min}(p) \leq f(p)$ for any f whatsoever is the full problem.

A Verification details

This appendix documents every computer-assisted claim of the paper (their proof status is stated in Remark 1.7): the finite enumerations that are *proof components* (Appendix A.1), the large-scale certificate computations behind Section 6 and how each dataset is verified (Appendix A.2), and the commands that reproduce all of it (Appendix A.3). All code is in the accompanying repository; every check is deterministic and uses exact integer arithmetic only.

A.1 Proof components

Each component below is an exhaustive enumeration of a finite space whose sufficiency is established in the text (Theorem 1.3 for the exact criteria; for Lemma 1.9, monotonicity of the reachability of the target in support and multiplicities, checked at minimal exponents). Each run completes in seconds on commodity hardware, except the largest support-bound check ($R = 107$: 331 s); the entry points named are in the module `erdos_straus.theory`.

Trusted computing base. The certificate data does not rest on the solver: every certificate claim is re-checked by evaluating the integer identity $4abc = p(bc+ca+ab)$ directly, so the trusted base for the data is a bignum multiply-and-compare (the host language's built-in integers), not the search pipeline. The finite case analyses trust more — the enumeration code itself, roughly sixty lines per check. Four mitigations apply. First, the enumerated spaces are defined by statements proved in the text (Theorem 1.3, Lemma 4.1), and the code mirrors those definitions one-to-one. Second, independent implementations agree where they overlap: the subset dynamic program and the direct binomial enumeration verify identical claims at $R = 19, 23$, and the exact criteria are cross-checked against ground-truth divisor search on up to 1,587,581 primes with zero discrepancies. Third, every check is deterministic, uses exact integer arithmetic only (no floating point, randomness, or external libraries), and runs from the repository in seconds (the largest, the $R = 107$ dynamic program, in 331 s) — an independent re-implementation is an afternoon's work, and we would welcome one.

Fourth, much of the development is formalized in the accompanying repository (`lean/`) in Lean 4 [3] with its mathematical library mathlib [10]. For readers unfamiliar with proof assistants: Lean is a language in which theorem statements and proofs are written formally and every proof is checked by a small trusted *kernel*, so a theorem accepted by the kernel is correct relative to its formal statement, independently of who — or what — wrote the proof. Two Lean-specific mechanisms matter below: `native_decide` discharges a decidable proposition by running compiled code, which extends the trusted base from the kernel to Lean's compiler and evaluator; and `#print axioms` reports, for any theorem, exactly which axioms (including any such evaluator appeal) its proof ultimately uses. The formalization comprises:

- *The elementary layer*, as ordinary kernel-checked proofs: certificate soundness and integrality, both directions of Theorem 1.2, both families of Proposition 1.12, Theorem 1.8, and the ladder's elementary layer — Lemmas 5.1 and 5.2 in full, and the selected-residual identity of Section 5 (the formalization shows the coprimality hypothesis of Lemma 5.1 is not needed: for $q \mid R$ both symbols vanish).

- *The enumeration layer*: the reachability model underlying Theorem 1.3 is defined in the multiplicative structure of $\mathbb{Z}/R$ directly, and both halves of Lemma 4.1 are proved as theorems, with no computational content to trust. The finite checks behind Theorem 1.4 (1,536 configurations, checked as the 384 that remain after the neutral class collapses), Theorem 1.5 (497,664 capped configurations, including the exact two-case failure classification), and Lemma 1.9 at $R = 19, 23$ (437,580 resp. 7,759,752 checks) are discharged by `native_decide`.

- *The coordinate bridges*: the larger enumerations are *evaluated* in discrete-log coordinates for speed, but the translation to the multiplicative model is itself a proved theorem, so the coordinates add nothing to the trusted base; the $R = 11, 19, 23$ statements hold in the multiplicative model, each reporting exactly its enumeration's single evaluator axiom.

- *The reduction of Theorem 1.3*: membership of the target class in the reachability model is proved equivalent to the existence of a divisor $k \mid m^2$ in that class (via the fundamental theorem of arithmetic), and produces the explicit positive certificate (k, b, c) solving (1.1)

— closing the loop between the abstract model the enumerations check and the integer certificates the conjecture asks for.

- *Two capstones*: a composed corollary (if $(p + 19)/4$ has nine prime factors in pairwise-distinct unit classes other than 1 modulo 19, an explicit solution of (1.1) at p exists, machine-checked end to end), and Lemma 1.9 at $R = 31$ by *certified dynamic programming* — the $\binom{29}{15} = 77{,}558{,}760$ supports are never enumerated; a machine-checked induction shows the dynamic program covers every support, and the compiled evaluator runs the dynamic program itself together with the final check that every state realized by ≥ 15 classes hits all 30 targets (3,001 states in the run) — the program's *correctness* is proved; its evaluation is what `native_decide` trusts.

Every main theorem is audited with `#print axioms`, and helper lemmas are covered transitively through them. (Not formalized: Theorem 1.6, Theorem 1.10, Propositions 1.1, 2.1 and 2.2, and the analytic arguments of Sections 4 and 5 — these rest on the Python layer and the hand proofs alone.) For these enumerations the sixty-line trust base above is thus replaced by the Lean toolchain evaluating a formally stated proposition — and the Python and Lean implementations, written independently against the same statements, agree. (The development notes and Lean sources use working letter names for several results; the repository's documentation includes the dictionary.)

In addition (as sanity checks, not proof components), the criteria of Theorems 1.2, 1.4, 1.5 and 1.6 were compared against ground-truth divisor search on the complete solvability data below 10^9: agreement in 1,587,581/1,587,581 cases for $R = 3$, 41,421/41,421 sampled failure cases for $R = 7$ (all Type I, as Theorem 1.4 requires), 158,759/158,759 sampled cases for $R = 15$, and 79,380 sampled cases for $R = 11$, with zero discrepancies; the general composite-R engine (`solvable_exact_general`) likewise agrees on 7,938 sampled primes for every composite admissible residual, and the strong form of Theorem 1.10(ii) is confirmed numerically at every residual checked (`verify_support_bound_strong`). One check named in this paragraph's vicinity is *not* a sanity check: `kneser_support_general` is a proof component of the $R \leq 107$ clause of Theorem 1.10(iii), and is listed as such in Table 5. The identity of Theorem 1.8 and its forced-prime corollary were likewise checked on 135,588 resp. 61,438 instances from the same data, with zero violations (Corollary 2.3).

A.2 Certificate datasets and their verification

The computational claims of Section 6 rest on the following datasets, all archived in the repository under `data/`. A certificate (R, a, b, c) for p is *accepted* only if $4abc = p(bc + ac + ab)$ holds in exact integer arithmetic and $R = 4a - p$; for the compact minimal-R maps, the verifier *reconstructs* the triple from (p, R) by the divisor search of (1.2) and then applies the same exact check, so a map entry is verified as a genuine certificate, not taken on faith.

The masks dataset of Table 6 is the basis for the covering-set statements (the 18-residual cover, the lower bound 12, the four critical primes), the failure taxonomy, and the calibration of Section 6; the minimal-R maps are the basis for Table 3, the record, and the gap analysis.

A.3 Reproduction

With the repository installed (`pip install -e .`), the commands of Table 7 regenerate and re-verify everything above. Generation of the full 10^{10} map takes about 11 minutes on four cores; every other command runs in seconds to minutes.

The ladder census of Section 5 is reproduced by `python -m erdos_straus.burgess_scan` (the 10^9 population) and by the scaled drivers in the same module for the window samples and the deep tail; the archives are the five `burgess_*.json` files under `data/analysis/`.

The full test suite (`python -m pytest`; 56 tests) exercises the proof-component checks, the exact criteria against ground truth, and sampled dataset validation on every run. Table 8 summarizes the computational environment.

Data and code availability

All code, all datasets of Table 6, and the Lean development are openly available in the repository `https://github.com/jeffmhopkins/Erd-s-Straus-attack`.

result	proof inputs	proved in	machine-checked?
Prop. 1.1	elementary	§1.1	—
Thm. 1.2 (R=3)	elementary	§3	Lean, both directions
Thm. 1.3	elementary	§3	model + reduction in Lean
Thm. 1.4 (R=7)	Prop. 2.1 + Thm. 1.3 + enumeration (1,536)	§3	Lean (`native_decide`)
Thm. 1.5 (R=11)	Thm. 1.3 + enumeration (497,664)	§3	Lean (`native_decide`)
Thm. 1.6 (R=15)	Thm. 1.3, composite form + enumeration (349,920)	§3	Python only
Thm. 1.8	quadratic reciprocity	§2	Lean
Lem. 1.9	Lem. 4.1 + enumerations / DP	§4	Lean at $R = 19, 23, 31$
Thm. 1.10	Kneser's theorem + Lem. 4.1; (iii) upgrade at $R \leq 107$: enumeration	§4	Python, (iii) enumeration only
Thm. 1.11	Thms. 1.2, 1.4–1.6, 1.10 + Fundamental Lemma	§4	no
Prop. 1.12	elementary + Thm. 1.11	§4	families in Lean
Thm. 1.13	Prop. 2.2 + Fundamental Lemma	§4	no
Thm. 4.5	Thm. 1.10(iii), general tier + Brun's sieve, uniform constants	§4	no
Lems. 5.1–5.2	Jacobi reciprocity, elementary	§5	Lean
Thms. 5.3–5.5	Thm. 1.10 / Prop. 2.2 + sieve	§5	no
Thm. 5.7	Prop. 2.1 + half-dim. sieve [8]	§5	no
Thm. 5.9	Hypothesis 5.8 (measured) + large sieve	§5	no (conditional)

Table 1: Audit map: what each main result depends on, where it is proved, and its formal-verification status. "—" means the result is elementary enough that no machine check was warranted; "no" means the proof is by hand only. Theorems 5.5 and 5.7 are established in outline (proof sketches).

population	first-rung success	ladder ($R \leq 400$)
all hard $p < 10^9$ (exhaustive)	94.78%	100% (two q's)
10^9–3.2×10^9 (sampled)	95.4%	100%
3.2×10^9–10^{10}	95.0%	100%
10^{10}–3.2×10^{10}	95.8%	100%
3.2×10^{10}–10^{11}	95.7%	100%
10^{11} deep tail ($R_{\min} \geq 43$, complete)	75.1%	99.99% (100% at $R \leq 435$)

Table 2: The selected-residual census: success rates at the first rung and along the q-ladder, capped at rungs $R_j \leq 400$. "(two q's)": below 10^9, exactly two primes needed the ladder of the *second*-least non-residue to resolve within the cap. The deep-tail row conditions on $R_{\min} \geq 43$, the adversarial 0.016% of the 10^{11} population; there, raising the cap to $R \leq 435$ restores 100%. The per-rung failure rate is flat in p across two decades, including on the deep-tail population.

R	count	R	count	R	count
3	69,951,190	39	25,258	75	29
7	31,335,473	43	6,850	79	50
11	19,439,771	47	8,811	83	23
15	3,974,286	51	1,450	87	8
19	2,441,687	55	1,369	91	3
23	1,070,808	59	954	95	5
27	187,963	63	283	99	1
31	193,848	67	104	103	1
35	30,784	71	209	107	1

Table 3: Distribution of the minimal residual $R_{\min}$ over the 128,671,219 hard primes below 10^{11} (plotted in Figure 1).

bin ($p \sim$)	10^6	10^7	6×10^7	2.5×10^8	10^9
density $\times \log x$	5.14	5.19	5.18	5.17	5.16

Table 4: The proportion of hard primes near x that fail both $R = 3$ and $R = 7$, multiplied by $\log x$, in five bins spanning three decades of the 10^9 data (x is the bin's representative value). Constancy of the product is the decay rate predicted by the base case of Theorem 1.11.

component	configuration space	count	entry point
Thm. 1.4 ($R = 7$)	class multisets in $(\mathbb{Z}/7)^*$, multiplicities ≤ 3, with the forced even factor, $p \bmod 7 \in \{1, 2, 4\}$, and the consistency relation	1,536	`verify_R7_finite`
Thm. 1.5 ($R = 11$)	dynamic program over $(\mathbb{Z}/11)^*$ with the forced factor 3; collapses to 25 equivalence states (16 succeed, 6 fail by character, 3 by budget)	25	`finite_criterion_dp`
Thm. 1.6 ($R = 15$)	class multisets in $(\mathbb{Z}/15)^*$, multiplicities ≤ 5, with the forced even factor, $p \bmod 15 \in \{1, 4\}$, and the consistency relation (349,380 succeed, 540 fail all-residue, zero budget failures)	349,920	`verify_R15_finite`
Lem. 1.9 ($R = 19$)	all supports of size 9 over the nonzero logs $\times$ all classes of p, at minimal multiplicities	437,580	`verify_support_bound`
Lem. 1.9 ($R = 23$)	all supports of size 11, likewise	7,759,752	`verify_support_bound`
Lem. 1.9 (R = 19–107, all twelve primes)	subset dynamic program over reachability-mask states; state counts 271 (R=19), 3,001 (R=31), 2,414,145 (R=83), 19,530,971 (R=107); zero violations throughout	$\approx$ 19.5M	`verify_support_bound_dp`
Thm. 1.10(iii), $R \leq 107$ clause	maximal support of a proper product-set in $(\mathbb{Z}/R)^*$ equals $\varphi(R)/2 - 1$ for every composite admissible $R \leq 107$ (e.g. $R = 95$: max 35 over 583,455 realizable product-sets)	12 moduli	`kneser_support_general`

Table 5: Machine-verified proof components: enumerated configuration spaces and the entry points in `erdos_straus.theory`.

dataset	entries	verification performed
explicit triples, $p < 1.2 \times 10^8$	213,131	every triple checked exactly
minimal-R map, $p < 10^9$	1,587,581	*exhaustive*: every entry reconstructed and checked exactly
minimal-R map, $p < 10^{10}$	14,215,707	143,000 systematically sampled entries reconstructed and checked; additionally *all* 842 tail entries ($R \geq 51$) checked with full minimality (no smaller admissible R succeeds)
solvability masks, $p < 10^9$	$27 \times 1{,}587{,}581$	per-prime solvability of every admissible $R \leq 107$ by full divisor-class computation; consistency with the minimal-R map
R-sequence dataset, $p < 10^{11}$	128,671,219	257,343 systematically sampled entries reconstructed and checked; *all* 20,151 tail entries ($R \geq 43$) checked with full minimality; prime array pinned by sha256; pipeline validated byte-for-byte against the exhaustively verified 10^9 map

Table 6: Certificate datasets and the verification applied to each.

claim	command
Thm. 1.4 finite check	python -c "from erdos_straus.theory \ import verify_R7_finite; \ print(verify_R7_finite())"
Thm. 1.5 enumeration	finite_criterion_dp(11, forced_logs=[_dlog_table(11)[1][3]]) from erdos_straus.theory (as in the test suite, tests/test_solver.py)
Lem. 1.9 at $R = 19, 23$	verify_support_bound(19), (23)
Thm. 1.6 finite check	python -c "from erdos_straus.theory \ import verify_R15_finite; \ print(verify_R15_finite())"
Thm. 1.10 numerical confirmation	verify_support_bound_strong(31), kneser_support_general(15)
generate the 10^{10} map	python -m erdos_straus.bulk_generate \ -max 10000000000 -segmented -store rmap
verify a dataset exhaustively	python -m erdos_straus.verify data/<file>
distribution / covering / taxonomy	python -m erdos_straus.analyze dist \ -rmap data/hard_primes_1e9_minimalR.json.gz (also cover, tail); python -m erdos_straus.theory
Lean formalization (elementary + enumeration layers)	cd lean/ErdosStraus lake exe cache get && lake build

Table 7: Reproduction commands. Shell commands are given verbatim, with \ marking an explicit line continuation within a single command; the finite-check rows are Python calls from the module erdos_straus.theory.

implementation	Python 3, exact integer arithmetic only; no external libraries in any proof-component check (`sympy` appears only in the reference solver)
factorization	trial division by sieved primes up to $\sqrt{n}$ (numpy-sieved prime tables for bulk generation)
wall-clock times	10^{11} generation: 2.85 h on 4 cores; 10^{10}: 656 s; 10^9: 49 s; largest proof-component check ($R = 107$ dynamic program): 331 s
verification	every certificate re-checked via $4abc = p(bc+ca+ab)$; exhaustive below 10^9; 257,343 samples plus all 20,151 tail entries at 10^{11} (Table 6)
integrity	datasets archived in the repository; the 10^{11} prime array and R-sequence are pinned by SHA-256 in `hard_primes_1e11_minimalR.meta.json` (prefixes `53cf17c4...`, `2a69f3aa...`)
formal layer	Lean 4.32.2 + mathlib (pinned toolchain); five `native_decide` checks; every main theorem audited by `#print axioms`

Table 8: Computational environment and integrity summary.

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